National University of Sciences and Technology
College of E&ME (Department of Mechatronics)
Midterm Exam Practice Problems(DE-46 MTS)

Subject Code: ME-474
Date: Mar 4, 2026
Time Limit: – hours
Instructor: Usman Asad

Subject: Fundamentals of Thermal Sciences
Timing: – hrs
Max Marks: –

Instructions:
This document contains 14 pages. These are a collection of previous exam and assignment questions over the years, based on the material already covered in the course

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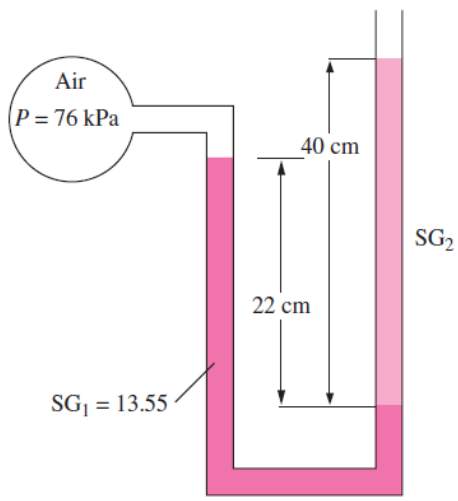
1. (5 points) Mark box if true.
   - ◯ The units of specific heat is kJ/kg [False]
   - ◯ Density of a fluid is an extensive property [False]
   - ◯ An Open System is also called a Control Volume [True]
   - ◯ Heat transfer is equal to Work done in an adiabatic process [False]
   - ◯ If the pressure and Temperature of water is known, we can determine its specific volume [True]

2. (10 points) Fill in the blanks
   (a) The gage pressure of tank is 20 psi. The absolute pressure will be **34.7 psi**
   (b) A 4 kW electric heater is kept on in room for 15 mins. The amount of energy transfered to the room is **3600** kJ
   (c) A mercury barometer is used to measure atmospheric pressure. The height of the mercury column will be around **760 mm / 30 in** The specific weight of mercury is $\lambda = \rho g = 0.49\ lb/in^3 = 133.7\ kN/m^3$ and $1\ atm = 14.7\ psi = 101.325\ kPa$
   (d) $45^o\ C$ is equal to **113** Fahrenheit
   (e) The SI unit of thermal conductivity 'k' is W/mK
   (f) 150 lbf converted in to SI units is **667.2 N**
   (g) 700 N converted in to kgf is **71.4 kgf**
   (h) The vacuum pressure of tank is 5 psi. The absolute pressure will be **9.7 psi**
   (i) 273 K is equal to **32** Fahrenheit
   (j) The SI unit of the Stefan Boltzmann constant 'σ' is W/m^2K^4

3. (10 points) Consider a double-fluid manometer attached to an air pipe shown in figure below. If the specific gravity of one fluid is 13.55, determine the specific gravity and density of the other fluid for the indicated absolute pressure of air. Take the atmospheric pressure to be 100 kPa



Solution:

$$P_{atm} + \rho_2 g h_2 - \rho_1 g h_1 = P_{air}$$

$$\rightarrow 100000 + \rho_2 \times 9.8 \times 0.4 - 13550 \times 9.8 \times 0.22 = 76000$$

Solving for ρ_2, gives us $\rho_2 = 1330\ kg/m^3$, so $SG_2 = 1.33$
Note: This is Problem 1-76 from the Cengel and Boles textbook.

4. (30 points) Consider 2 kg of air in a Piston-Cylinder assembly with the following processes in a thermodynamic cycle:

- **Process** $1 \rightarrow 2$**:** Adiabatic compression where $P_1 V_1^n = P_2 V_2^n$, with $n = 1.385$ from atmospheric pressure and $V_1 = 3\ m^3$to a final volume of $V_2 = 1\ m^3$
- **Process** $2 \rightarrow 3$**:** Isothermal expansion to back to atmospheric pressure
- **Process** $3 \rightarrow 1$**:** Isobaric compression

Draw the cycle on a P-V diagram. **Evaluate** the work done in each process. **Evaluate** the heat transfer for each process. **Estimate** the thermodynamic efficiency of the cycle.

Note that for a polytropic process $PV^n = const$:

$$n = 0 \rightarrow W_{12} = P(V_2 - V_1)$$

$$n = 1 \rightarrow W_{12} = P_1 V_1 \ln(\frac{V_2}{V_1})$$

Otherwise:

$$W_{12} = \frac{P_2 V_2 - P_1 V_1}{1 - n}$$

Use the following constants for Air
$R = 0.287\ kJ/kgK \quad ; \quad c_p = 1.033\ kJ/kgK \quad ; \quad c_v = 0.7457\ kJ/kgK$

Solution:
Taking atmospheric pressure to be 100 kPa, pressure at 2 is given by:

$$P_2 = P_1 (V_1/V_2)^n = 457.8\ kPa$$

Work done for 1 to 2 is:

$$W_{12} = \frac{P_2 V_2 - P_1 V_1}{1 - n} = -410.2 kJ$$

Process 2 to 3 is isothermal. Volume at 3 is given by:

$$P_2 V_2 = P_3 V_3 \rightarrow V_3 = (P_2/P_3) V_2 = 4.58\ m^3$$

Work done for the isothermal process (n=1) is:

$$W_{23} = P_2 V_2 \ln(\frac{V_3}{V_2}) = 696.7\ kJ$$

Work done for for process 3 to 1:

$$W_{31} = P(V_1 - V_3) = -157.9\ kJ$$

Before we calculate heat transfer, we need the temperatures. At point 1:

$$P_1V_1 = mRT_1 \rightarrow T_1 = 249.4^o\ C$$

$$P_2V_2 = mRT_2 \rightarrow T_2 = 524.5^o\ C$$

and $T_2 = T_3$ because its an isothermal process Heat transfer for process 1 to 2 is zero because its an adiabatic process Heat transfer for process for process 2 to 3 is equal to the work done, because of the 1st law:

$$\Delta U_{23} = Q_{23} - W_{23}$$

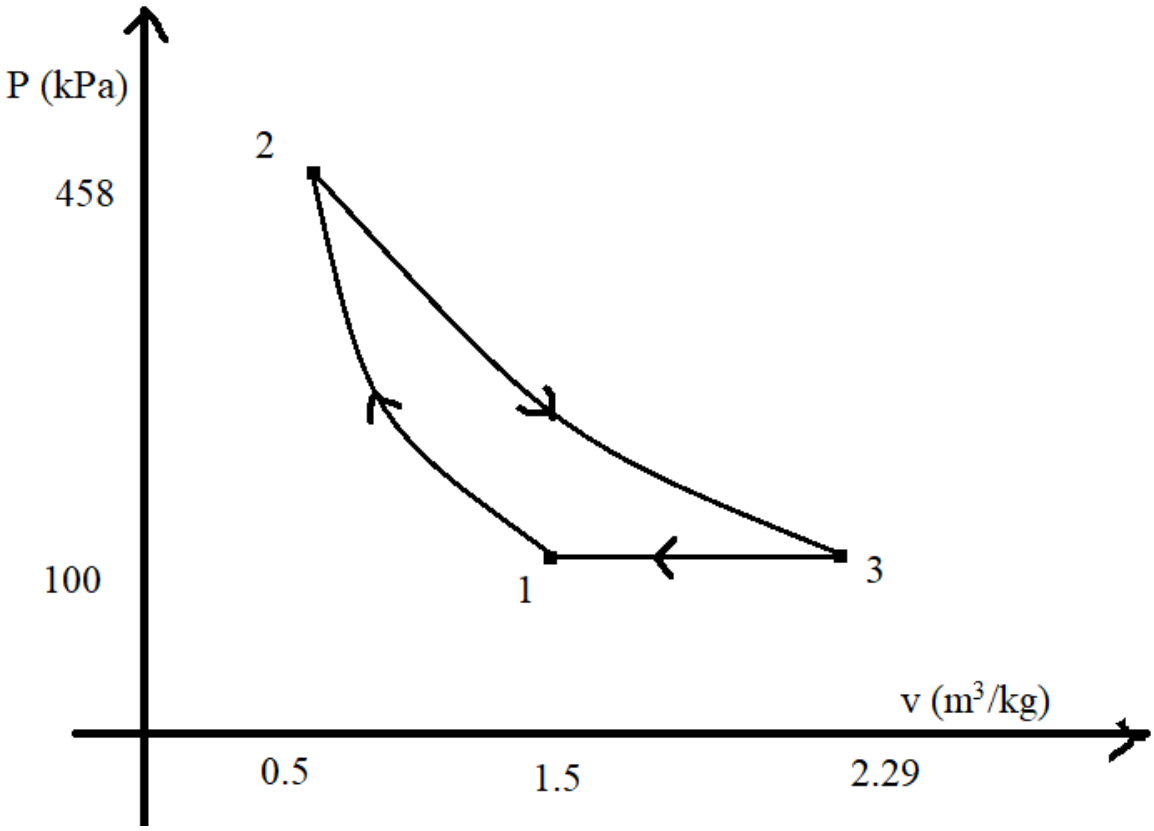
For $\Delta U_{23} = 0$ because temperature at 2 and 3 are the same, we have $= Q_{23} = W_{23} = 696.7\ kJ$. This is Q_{in}
Process 3 to 1 is a constant pressure process, so:

$$Q_{31} = mc_p(T_1 - T_3) = -568.1\ kJ$$

So $Q_{out} = 568.1\ kJ$. Finally efficiency is:

$$\eta = 1 - \frac{Q_{out}}{Q_{in}} = 0.1845\ or\ 18.45\%$$

The Pv diagram:

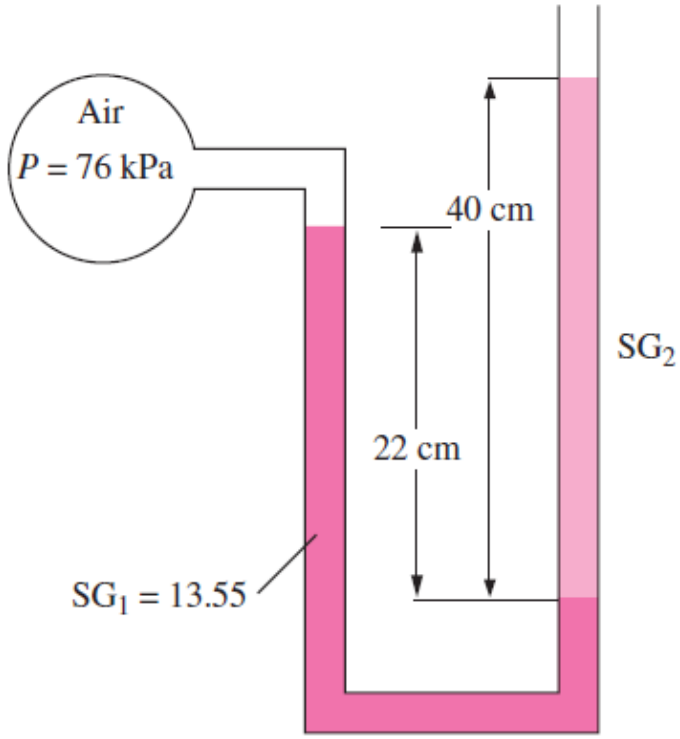


Note: PV diagram looks exactly the the same. Just multiply the values of specific volume given the bottom with m=2

5. (10 points) Ambient air at a temperature of 25°C and a pressure of 100 kPa in a piston cylinder assembly is compressed (in an adiabatic, reversible process) from a volume of 78 cm^3 to 8.2 cm^3. Estimate the mass of the air. What will be the temperature and pressure of air after compression? How much work will be required for this process?

6. (10 points) An average household consumes around 360 kWh of electrical energy per month. Assuming all that energy came from Natural Gas Power plants having an efficiency of around 60%. In order to judge the environmental impact of an average household, estimate the CO_2 emissions due to one household, in kg/year. Assume that during combustion of natural gas, around 50 kg of CO_2 are produced per 1 million kJ of heat released.

7. (10 points) A gasoline line is connected to a pressure gage through a double-U manometer, as shown in the figure. If the reading of the pressure gage is 370 kPa, determine the gage pressure of the gasoline line.



Solution:

$$P_{petrol} + \rho_{petrol} g(0.22) + \rho_{Hg} g(0.1) - \rho_{Oil} g(0.50) + \rho_{H2O} g(0.45) = 370,000$$

$$P_{petrol} = 354.6 kPa$$

8. (25 points) Consider air in a Piston-Cylinder assembly with the following processes in a thermodynamic cycle:

   - **Process** $1 \rightarrow 2$**:** Adiabatic compression where $P_1 V_1^k = P_2 V_2^k$, with $k = 1.35$ from $P_1 = 100\ kPa$, $V_1 = 522.7\ cm^3$, $T_1 = 333K$ to $V_2 = 22.7\ cm^3$, $T_2 = 996.5K$
   - **Process** $2 \rightarrow 3$**:** Constant Pressure expansion to $V_3 = 79.7\ cm^3$
   - **Process** $3 \rightarrow 4$**:** Adiabatic expansion where $P_3 V_3^k = P_4 V_4^k$, with $k = 1.35$ to $V_4 = 522.7\ cm^3$
   - **Process** $4 \rightarrow 1$**:** Constant Volume Process

   Draw the cycle on a P-V diagram. **Estimate** the mass of air in the cylinder. **Evaluate** the work done in each process. **Estimate** the thermodynamic efficiency of the cycle, give by:

$$\eta = \frac{W_{cycle}}{Q_{23}}$$

Solution:
The mass of the air in the cylinder is given by:

$$m = \frac{P_1 V_1}{R T_1} = \frac{100 \times 522.7 \times 10^{-6}}{0.287 \times 333} = 0.55 \times 10^{-3}\ kg$$

Note that the mass doesn't change. You can use values at any state.

Pressure after process $1 \rightarrow 2$ is given by:

$$P_2 = P_1 (\frac{V_1}{V_2})^k = 100 \times (\frac{522.7}{22.7})^{1.35} = 6902\ kPa$$

Work done in Process $1 \rightarrow 2$:

$$W_{12} = \frac{P_2 V_2 - P_1 V_2}{1 - k} = -298\ J$$

Work done in Process $2 \rightarrow 3$:

$$W_{23} = P_2 (V_3 - V_2) = 393.4\ J$$

Using the same equations as before, Pressure $P_4 = 544.9\ kPa$, and Work done in Process $3 \rightarrow 4$: $W_{34} = 758\ J$

There is no work done in Process $4 \rightarrow 1$
Net Work done is the cycle is: $W_{cycle} = W_{12} + W_{23} + W_{34} = 853.1\ J$

Heat added for Process $2 \rightarrow 3$ is given by:

$$Q_{23} = \Delta H = m c_p \Delta T = m c_p (T_3 - T_2)$$

The temperature at 3 is given by (using the Ideal Gas Law):

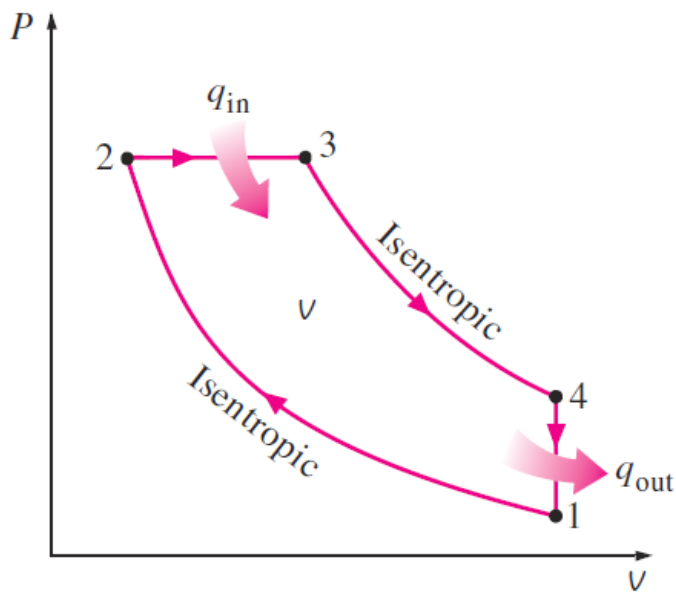
$$T_3/V_3 = T_2/V_2 \rightarrow T_3 = 3500K$$

$$\rightarrow Q_{23} = 2724\ J$$

Finally, the efficiency is:

$$\eta = \frac{W_{cycle}}{Q_{23}} = 31.32\%$$

The PV diagram of the cycle will essentially look like this:

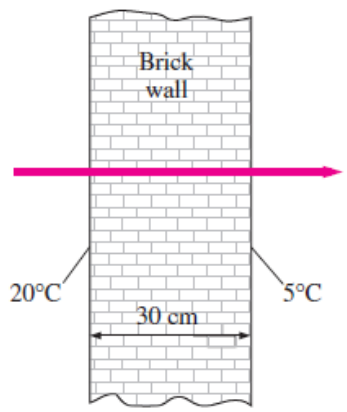


This cycle is the called the Air-Standard Diesel Cycle. The problem we solved is an approximation for one of the pistons of the Toyota Corolla 2.0D

9. A piston–cylinder device contains helium gas initially at 150 kPa, 20°C, and 0.5 m3. The helium is now compressed in a polytropic process (PV^n =constant) to 400 kPa and 140°C. Determine the heat loss or gain during this process.
   Answer: 11.2 kJ loss

10. A 0.5 m^3 rigid tank contains refrigerant-134a initially at 200 kPa and 40 percent quality. Heat is transferred now to the refrigerant from a source at 35°C until the pressure rises to 400 kPa. Determine (a) the entropy change of the refrigerant, (b) the entropy change of the heat source, and (c) the total entropy change for this process.
    Answers: (a) 3.880 kJ/K, (b) 3.439 kJ/K, (c) 0.441 kJ/K

11. The inner and outer surfaces of a $5m \times 6m$ brick wall of thickness 30 cm and thermal conductivity 0.69 W/m°C are maintained at temperatures of 20°C and 5°C, respectively. Determine the rate of heat transfer through the wall, in W.



12. (10 points) A power plant, with a power output of 400 MW, initially had an efficiency of 35%. The efficiency of the plant was increased to 38% through design changes. Coal is used a the fuel source in the boiler. During combustion of coal, around 95 kg of CO_2 are produced per 1 million kJ of heat released. Estimate the environmental impact of this increased efficiency through savings in CO_2 emissions, in tons per year.

13. (16 points) 2 kg of air is compressed in a piston-cylinder assembly in a polytropic process (Pv^n = constant) with $n = 1.4$. No heat transfer takes place during the process
The initial state of the Air is $P_1 = 7\ bar$; $T_1 = 180°C$; $V_1 = 0.1m^3$; $u_1 = 325\ kJ/kg$.

- Compute the work done and u_2 if the air is compressed to $V_2 = 0.01\ m^3$
- Draw the process on a P-v diagram

Solution: Pressure P_2 may be computed using $P_1V_1^n = P_2V_2^n$

$$P_2 = P_1(V_1/V_2)^n = 7 \times (10)^{1.4} = 175.8\ bar$$

Work done W_{12} is given by:

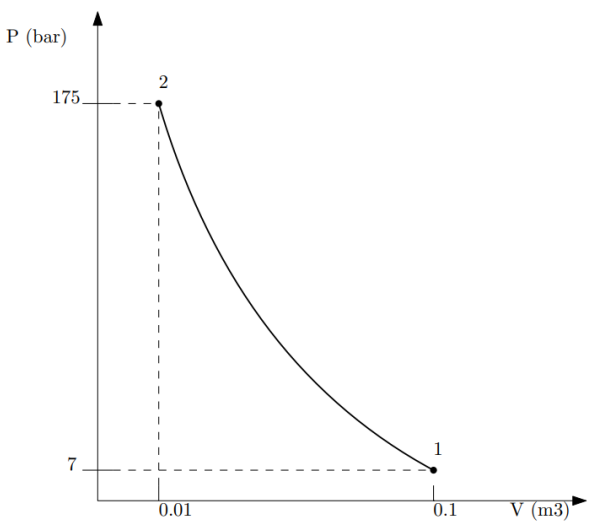
$$\rightarrow W_{12} = \frac{P_2V_2 - P_1V_1}{1-n} = \frac{175.8 \times 0.01 - 7 \times 0.1}{1 - 1.4} \times 100 = -264.6kJ$$

The negative sign indicates work is done **on** the system. The specific internal energy u_2

may be found using the first Law of thermodynamics:

$$U_2 - U_1 = m_2 u_2 - m_1 u_1 = Q_{12} - W_{12}$$

$$\rightarrow u_2 = u_1 + \frac{W_{12}}{m} = 325 + 264.6/2 = 457.4\ kJ/kg$$



14. (12 points) 1kg of water at atmospheric pressure is heated in a constant pressure process from 25^oC to 250^oC. Subsequently, the water is cooled in a constant volume process to saturated steam.

- Estimate the Temperature of the final state
- Draw the process on a P-v and T-v diagram

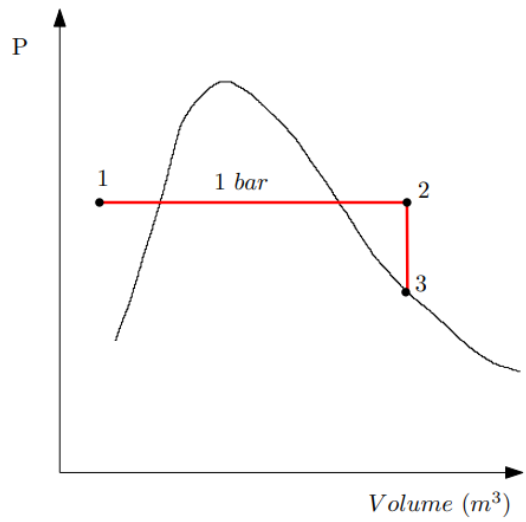
Solution:

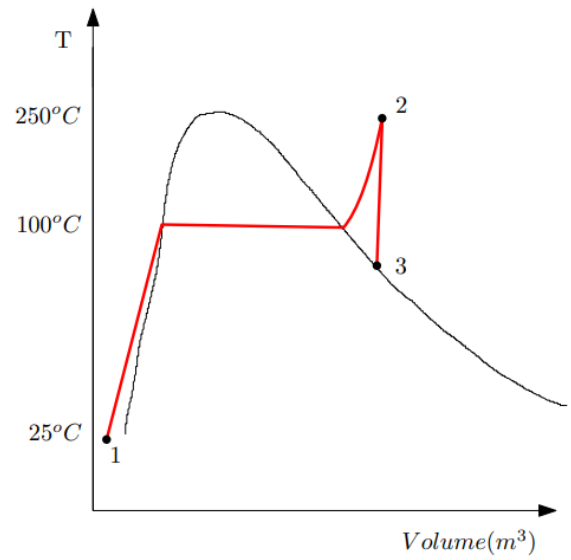
Water is in superheated state at 250^oC and 1 bar. The specific volume at state 2 is $2.4m^3/kg$ using Table A-4. More exactly, using interpolation:

$$v_2 = 2.359 + (2.546 - 2.359) \times \frac{1}{4}\ m^3/kg$$

Note that process $2 \rightarrow 3$ is a constant volume process, so $v_2 = v_3$. So we have to find (or estimate) a temperature in table A-3 or A-2, such that the specific volume of saturated

vapor, v_g is equal to v_3. The closest match is at $P = 0.7\ bar$ and $T = 89.95^oC$, with $v_g = 2.365$. So $T_3 \approx 89^oC$. More precisely, $T_3 = 89.48^oC$





15. (6 points) Determine the heat transfer rate per unit area for a 25 cm thick concrete wall with a temperature of 10°C on one end and a temperature of 26°C on the opposite end. The thermal conductivity for concrete is 0.7 W/(m K)

Solution:

$$\frac{\dot{Q}_x}{A} = -k\frac{dT}{dx} = -0.7 \times \frac{26-10}{0.25} = -44.8\ W/m^2$$

16. (6 points) Find the quality of a saturated liquid-vapor mixture of water at $T = 100°$C and $u = 1800kJ/kg$

Solution:

The specific internal energy is given by:

$$u = u_f + xu_{fg}$$

$$\rightarrow x = \frac{u - u_f}{u_g - u_f} = \frac{1800 - 418.94}{2506.5 - 418.94} = 0.661$$

17. (4 points) Find the Volume of 7.6 kg of Saturated Steam at 900 kPa pressure

Solution:

Using Table A-3, at 9 bar: $v = 0.2150m^3/kg$ So Volume $V = 7.6 \times 0.2150 = 1.634\ m^3$

18. (3 points) A mercury barometer is used to measure atmospheric pressure. Estimate the height of the mercury column in inches, given the specific weight of mercury $\lambda = \rho g = 0.49\ lb/in^3 = 133.7\ kN/m^3$ and $1\ atm = 14.7\ psi = 101.325\ kPa$

    Solution:

$$\rho g H = 14.7\ psi \rightarrow H = 14.7/0.49 = 30\ inch$$

19. (3 points) Find the Enthalpy of Vaporization (Latent heat of vaporization) of water at 10 bar

    Solution:

    Using Table A-3 at 10 bar, Enthalpy of Vaporization:

$$h_{fg} = 2015.3\ kJ/kg$$

20. (12 points) In an automobile, the tires are pressurized with the gage pressure set at 30 pounds per square inch gage (psig) when the temperature of air is 30°C. The temperature drops to 8°C. What will be the gage pressure in the tires?

21. (10 points) Air in a piston-cylinder assembly undergoes isothermal expansion. i.e. A polytropic process (Pv^n = constant) with n=1.
    The initial state of the Air is $P_1 = 7\ bar$; $V_1 = 0.1m^3$

    - Compute the work done if the air expands to $V_2 = 0.5\ m^3$
    - Determine the heat transfer during the process
    - Draw the process on a P-v diagram

22. (18 points) 2 kg of water undergoes the following processes:

    - **Process 1 to 2:** Constant Pressure Heating at 10 bar from 80°C to saturated vapor state.
    - **Process 2 to 3:** Constant volume cooling with a heat transfer of -300 kJ. (Heat removed from the system)

    Determine the Specific Volume (v), Specific Internal Energy (u), and Quality (x) at the final state. Draw the processes on a T-v and P-v diagram

23. (10 points) Compute the work done in expansion of water initially in the mixed region with $x = 0.4$ to saturated vapor at a constant pressure of 6 bar.

24. (10 points) In an automobile, the tires are pressurized with the gauge pressure set at 30 pounds per square inch gauge (psig) when the temperature of air is 30°C. The temperature drops to 8°C.

(a) Estimate the gauge pressure in the tires in psig?

(b) If the volume of the tire is 10 liters, estimate the mass of air that must be added to the tire to bring the pressure in the tire back to 30 psig (1 liter = 1000 cm^3)

25. (10 points) One kg of water vapor is heated in a closed tank at constant pressure from saturated vapor at 180°C to a final temperature of 350°C. Determine the initial and final volume and sketch the process on T–v and p–v diagrams.

26. (10 points) Ammonia enters a control volume operating at steady state at $p_1 = 14$ bar, $T_1 = 28$°C, with a mass flow rate of 0.5 kg/s. Saturated vapor at 4 bar leaves through one exit, with a volumetric flow rate of 1.036 m^3/min, and saturated liquid at 4 bar leaves through a second exit. Determine

    (a) the minimum diameter of the inlet pipe, in cm, so the ammonia velocity does not exceed 20 m/s.

    (b) the volumetric flow rate of the second exit stream, in m^3/min.

27. (18 points) Air in a piston-cylinder assembly initially has a volume of 50 cm^3 at $T_1 = 760K$ and $P_1 = 2400$ kPa. The specific heat at constant pressure, may be assumed to have a constant value of $c_p = 1.108$ kJ/kg K

    - Process 1 → 2: The air undergoes a constant volume heat addition to $T_2 = 1500K$. Compute the mass of the air, the heat added (in kJ) and the Pressure P_2 (in kPa)
    - Process 2 → 3: The air undergoes a constant pressure heat addition of 0.167 kJ. Compute the final pressure P_3 and volume V_3
    - Draw the process on a P-v diagram

    Solution:

    Process 1 → 2:

$$m = \frac{PV}{RT} = \frac{2400 \times 50 \times 10^{-6}}{0.287 \times 760} = 5.5 \times 10^{-4} kg$$

$$Q_{12} = m(u_2 - u_1) = mc_v(T_2 - T_1) = 0.3342\ kJ$$

$$\frac{P_2}{P_1} = \frac{T_2}{T_1} \rightarrow P_2 = \frac{P_1 T_2}{T_1} = 4737\ kPa$$

    Process 2 → 3:

$$Q_{23} = m(h_3 - h_2) = mc_p(T_3 - T_2)$$

$$\rightarrow T_3 = T_2 + \frac{Q_{23}}{mc_p} = 1774\ K$$

$$P_3 = P_2 = 4737\ kPa$$

$$\frac{V_3}{V_2} = \frac{T_3}{T_2} \rightarrow V_3 = 59.14\ cm^3$$

28. (12 points) Complete the below table, for Water (Values in Red are the answers):

T, °C	*P*, kPa	*u*, kJ/kg	Phase description
143.6	400	1450	**Saturated Liquid-Vapor Mixture**
220	**2318**	**2602**	Saturated vapor
190	2500	**805.3**	**Subcooled/Compressed Liquid**
466.6	4000	3040	**Superheated Vapor**

29. (18 points) Steam enters an insulated Turbine at $T = 360^o\ C$ and $P = 3$ MPa with a mass flow rate of 9.2 kg/s. 20% of the entering mass flow exits the turbine at 0.5 MPa and $180^o\ C$. The rest exits the Turbine from another location at 10 kPa and a quality of 90%. If the Inlet Velocity is 26.5 m/s, estimate the inlet piping diameter.
Solution:

$$\dot{m} = \frac{A_1 \overrightarrow{V_1}}{v_1} = \frac{\pi d^2}{4 v_1} \overrightarrow{V_1} = 9.2\ kg/s$$

Specific Volume at the inlet, from superheated table @ 3 MPa, 360°C:

$$v_1 = 0.0923\ m^3/kg$$

$$\rightarrow d = 0.202\ m$$

Mass flow rate at the first and second exit:

$$\dot{m}_2 = 0.2\dot{m}_i = 1.84\ kg/s$$

$$\dot{m}_3 = 0.8\dot{m}_i = 7.36\ kg/s$$